



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCBP401

CEMENT BASED BLOCKS AND PAVERS WORKS

Perform cement-based blocks and pavers works

Competence

RQF Level: 4

Learning Hours



80

Credits: 8

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 In Building Construction

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Purpose statement	This module describes the knowledge, skills, and attitudes required to make cement-based blocks pavers. It is intended to be pursued by learner at TVET level IV in Building Construction. At the end of this module, the learner will be able to: select tools, equipment and materials, prepare mortar, produce cement blocks and cement pavers, prepare ground base, and lay pavers.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative assessment			50%

Elements of competency and performance criteria

Elements of competency	Performance criteria
1. Select tools, materials and equipment	1.1. Tools are appropriately selected according to the work to be done.
	1.2. Equipment is appropriately selected according to the work to be done.
	1.3. Materials are adequately selected according to the work to be done.
2. Fabricate cement blocks	2.1. Materials are precisely quantified according to the mixing ratio.
	2.2. Materials are properly batched and mixed based on the design ratio.
	2.3. Mixed materials are correctly poured and compacted into the mould according to the designed shape.
	2.4. Cement blocks are adequately de-moulded and cured according to the standard.
3. Fabricate cement pavers	3.1. Materials are precisely quantified according to the mixing ratio.
	3.2. Materials are conveniently batched and mixed based on the design ratio.

	3.3. Mixed materials are correctly poured and compacted into the mold according to the designed shape.
	3.4. Pavers are adequately de-moulded and cured according to the standard.
4. Prepare ground base	4.1. The paving area is properly cleared and levelled in accordance to design.
	4.2. Ground base is accurately investigated according to the work specifications.
	4.3. The laying place is conveniently compacted in accordance to the type of traffic to be supported, and work specifications or desired compaction rate.
5. Lay pavers	5.1. The laying place is precisely set in accordance with design dimensions.
	5.2. Edge restraint is conveniently supported according to the standards.
	5.3. Sand bedding is evenly spread along the ground base.
	5.4. Pavers are correctly applied according to the design.
	5.5. Pavers' Joints are correctly filled according to the design/work specifications.
	5.6. Pavers' Joints are properly sealed according to the design/work specifications.
	5.7. Tools, equipment and work area are adequately cleaned according to standard.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify Types and size of raw materials ✓ Identify and describe cement mix proportion and its importance ✓ Identify various adhesives used in	✓ Prepare mixing batch ✓ Quantify and mix mortar ingredients ✓ Mould mortar ✓ Apply Compaction methods ✓ De-mold blocks or pavers ✓ Apply curing methods	✓ Use creativity and innovation throughout blocks and pavers fabrication and construction ✓ Pay attention to laying/construction patterns and details ✓ Demonstrate persistent, team

block work ✓ Identify water cement ratio ✓ Identify methods of curing of moulded cement blocks and pavers		working, flexibility, punctuality during blocks and pavers production and laying
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Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Select tools, materials and equipment. 2. Fabricate cement blocks 3. Fabricate cement pavers 4. Prepare ground base 5. Lay pavers
Learning outcome 1: select tools, equipment and materials	Learning hours: 5
Indicative content	
• Identification of tools ➤ Classification of tools ✓ Setting out tools ✓ Mixing tools ✓ Batching tools ✓ Cleaning tools ✓ Placing tools ✓ Moulding tools ✓ Levelling tools ➤ The uses of tools ➤ The maintenance of tools • Identification of equipment ➤ Classification of equipment ✓ Setting out equipment ✓ Cutting equipment ✓ Mixing equipment ✓ Batching equipment ✓ Cleaning equipment	

- ✓ Compaction equipment
- ✓ Transporting equipment
- ✓ Levelling equipment

➤ Uses of equipment

- **Identification of materials**

- Types of materials
- Properties of materials
- Factors for selecting cement blocks materials
- Factors for selecting pavers materials

Resources required for the learning outcome

Equipment	-Wheelbarrow, concrete mixers, truck mixers Steelhead hammer, Shovels, Sieves set.
Materials	-Cement, Fine aggregate, Coarse aggregate, Water, Stone, Bricks.
Tools	-Trowel, Spirit level, Squares, building line, Tape measure, roller brush, hand brush, scrapper, Tray, steel float, Sand paper, pan/bucket, mixing stick, strings (Road marking line), brushes, roller tray and bucket/pan and spade.
Facilitation techniques	• Brainstorming • Group discussion • Trainer guided
Formative assessment methods	• Written assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Essay • Open ended questions

Indicative content

- **Quantification of cement block materials**
 - Types of cement block materials
 - Specification of cement block materials
 - Methods of quantifying cement block materials
 - ✓ Manual method
 - ✓ Automatic/mechanical method
- **Batching of cement block materials**
 - Methods of batching
- **Mixing of cement block materials**
 - Mix ratio
 - Methods of mixing
 - Procedures of mixing
- **Pouring of cement mortar**
 - Process of pouring cement mortar into cement block mould
- **Compaction of cement-based blocks**
 - Advantages of compaction
 - Compaction method of cement-based blocks
 - ✓ Manual method
 - ✓ Mechanical method
 - Compaction process
- **De-moulding of cement-based blocks.**
 - De-moulding process
- **Curing of fabricated cement blocks**
 - Period of curing
 - Identification of drying types
 - ✓ Artificial drying
 - ✓ Natural drying
 - Application of cement block curing process complying with Rwanda standards

Resources required for the indicative content

Equipment	Wheelbarrow, concrete mixers, truck mixers, Steelhead hammer, Shovels, Sieves set...
Materials	Cement, Fine aggregate, Coarse aggregate, Water, Stone, Bricks...
Tools	Trowel, Spirit level, Squares, building line, Tape measure, roller brush, hand brush, scrapper, Tray, steel float, Sand paper, pan/bucket, mixing stick, strings (Road marking line), brushes, roller tray and bucket/pan, spade...
Facilitation techniques	• Demonstration with video aid • Simulation

	• Group discussion • Trainer guided
Formative assessment methods	• Performance assessment • Product based assessment • Matching questions • Multiple choice • Sentence completion • Short questions • Open ended questions

Learning outcome 3: Fabricate cement pavers.	Learning hours: 17
Indicative content	
• Quantification of cement pavers materials ➤ Types of cement paver materials ➤ Specification of cement pavers materials ➤ Methods of quantify cement pavers materials ✓ Manual method ✓ Automatic/mechanical method • Batching of cement pavers materials ➤ Methods of batching • Mixing of cement pavers materials ➤ Mix ratio ➤ Methods of mixing ➤ Procedures of mixing • Pouring of cement mortar ➤ Process of pouring cement mortar into cement pavers mould • Compaction of cement paver ➤ Advantages of compaction ➤ Compaction method ✓ Manual method ✓ Mechanical method ➤ Compaction process • De-moulding of cement pavers ➤ De-moulding process • Curing of fabricated cement pavers	

- Period of curing
- Identification of drying types
 - ✓ Artificial drying
 - ✓ Natural drying
- Application of cement pavers curing process complying with Rwanda standards

Resources required for the indicative content

Equipment	-Wheelbarrow, concrete mixers, truck mixers Steel head hammer, Shovels, Sieves set...
Materials	-Cement, Fine aggregate, Coarse aggregate, Water, Stone, Bricks...
Tools	-Trowel, Spirit level, Squares, building line, Tape measure, roller brush, hand brush, scrapper, Tray, steel float, Sand paper, pan/bucket, mixing stick, strings (Road marking line), brushes, roller tray and bucket/pan and spade...
Facilitation techniques	• Demonstration with video aid • Simulation • Group discussion • Trainer guided
Formative assessment methods	• Performance assessment • Product based assessment • Project based assessment

Learning outcome 4: Prepare ground base

Learning hours: 5

Indicative content

- **Preparation of working area**
 - Methods of assessing the paving area
 - Clearing and cleaning methods of the paving area
 - Types of waste
 - ✓ Biodegradable waste
 - ✓ Chemical waste
 - ✓ Demolition waste
 - ✓ E-waste
 - ✓ Industrial waste
 - ✓ Recyclable waste
- **Investigation of ground base moisture.**
 - Conduct soil investigation
 - Drainage system application
 - Watering of ground base
- **Excavation and levelling of the area to be paved**
 - Method of excavation
 - Techniques of levelling working area

• Compaction of working area ➤ Advantages of compacting ground base ➤ Methods of compacting ground base ✓ Manual method ✓ Mechanical method	
Resources required for the indicative content	
Equipment	-Wheelbarrow, concrete mixers, truck mixers Steel head hammer, Shovels, Sieves set...
Materials	-Cement, Fine aggregate, Coarse aggregate, Water, Stone, Bricks...
Tools	-Trowel, Spirit level, Squares, building line, Tape measure, roller brush, hand brush, scrapper, Tray, steel float, Sand paper, pan/bucket, mixing stick, strings (Road marking line), brushes, roller tray and bucket/pan and spade...
Facilitation techniques	• Demonstration with aid of video • Simulation • Trainer guided
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment

Learning outcome 5: Lay pavers	Learning hours: 35
Indicative content	
• Types of pavers ➤ Cement ➤ Concrete ➤ Brick ➤ Cobblestone ➤ Rubber ➤ Bluestone ➤ Flagstone ➤ Travertine • Methods/processes of Setting out the area to be paved ➤ The use of calibrated measuring tools and calibrated levelling tools ➤ Mark out the level of pavement • Supporting the edge of area to be paved ➤ Methods of supporting edge restraints complying with related Rwanda Standards ✓ Precast concrete kerbs ✓ Cut stone ✓ Plastic • Spreading the sand bedding along the ground base. ➤ Identification of the type of sand to be used	

- ✓ Coarse sand specification complying with Rwanda Standards
- ✓ Determination of thickness of sand layer applied

- **Laying of pavers on sand bedding surface**

- Identification of pattern:
 - ✓ Stack bond
 - ✓ Basket weave bond
 - ✓ Running bond
- Pavers laying method
 - ✓ Ridged laying method
 - ✓ Flexible laying method
- Construct pavers complying with Rwandan Standard: precast concrete paving blocks specification.

- **Filling of joints between pavers with fine grained sand**

- Types of joints
 - ✓ Mortar joint
 - ✓ Sand joint

- **Placing sealant in joint**

- Types of paver sealers
 - ✓ Solvent based paver sealer
 - ✓ Water based paver sealer
 - ✓ Wet look paver sealer
 - ✓ Gloss paver sealer
- Properties of sealant materials
 - ✓ Elasticity
 - ✓ Adhesion
 - ✓ Modulus
 - ✓ Compatibility
 - ✓ Weather ability
 - ✓ Cohesion

- **Cleaning work area, tools and equipment**

- Methods of clearing
 - ✓ manual clearing method
 - ✓ mechanized clearing method

- **Cleaning tools and equipment**

- Methods of cleaning
 - ✓ Manual cleaning
 - ✓ mechanized cleaning method

- **Storage of tools and equipment**

➤ Methods of storing tools and Equipment

Resources required for the indicative content

Equipment	-Wheelbarrow, concrete mixers, truck mixers Steel head hammer, Shovels, Sieves set...
Materials	-Cement, Fine aggregate, Coarse aggregate, Water, Stone, Bricks...
Tools	-Trowel, Spirit level, Squares, building line, Tape measure, roller brush, hand brush, scrapper, Tray, steel float, Sand paper, pan/bucket, mixing stick, strings (Road marking line), brushes, roller tray and bucket/pan and spade...
Facilitation techniques	• Demonstration with aid of video • simulation • Trainer guided
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment • Project based assessment

Integrated/Summative assessment

Integrated situation

MC located in Musanze district has the problem of a slippery pedestrian walkway and gate keeper house during the rainy season. In collaboration with Musanze district MC has gain a fund to make cement pavers for pedestrian way and cement blocks for construction of gate keeper house. JJY Company Ltd has won a tender of the above work and as a technician of Level 4 building construction, you are hired to complete the following tasks:

1. Calculate the number of cement pavers required
2. Calculate the number of cement blocks required
3. Produce the cement pavers within 8 hours
4. Produce the cement blocks within 8 hours
5. Cure all produced products in 3 hours
6. Lay cement paver for pedestrian way within 4 hours

All materials, tools and equipment are available on the site

Instructions:

- The Pedestrian way measurements are 6m *2m
- The building's wall measurements are 4m length and 3 m of height
- The size of paver is 5cm*15cm
- The size of block is 40cm*20cm*20cm and contain two connective holes
- The mixing ratio to be used for all tasks is: 1:3

Resources

Tools	-Trowel, Spirit level, Squares, building line, Tape measure, Roller brush, Hand brush, Scraper, Tray, Steel float, Sand paper, Pan/bucket, Extension pole, Mixing stick, Strings (Road marking line), Brushes, Paint Scraper, Roller, Roller tray
Equipment	PPEs, Compaction machine, Wheelbarrow, Vibrator machine, Concrete mixer, Gauge box, Mortar mixer, and Bucket/pan
Materials/ Consumables	Water, Sand, Cement, Oil, gravel, stone, Precasted concrete kerbs

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Select tools, equipment and materials, (10%)	1.1. Tools are appropriately selected according to the work to be done	Setting out tools are selected			1
		Batching and mixing tools are selected			0.5
		Placing tools are selected			0.5
		Moulding tools are selected			1
		Levelling tools are selected			0.5
		Cleaning tools are selected			0.5
	1.2. Equipment is appropriately selected according to the work to be done	Setting out equipment are selected			0.5
		Batching and mixing equipment are selected			0.5
		Levelling equipment is selected			1
		Compaction equipment are selected			1
		Transporting and placing equipment are selected			0.5
		Cleaning equipment are selected			0.5
	1.3. Materials are adequately selected according to the work to be done	Cement is selected			0.5
		Sand is selected			0.5
		Gravel is selected			0.5
		Oil is selected			0.5
Learning outcome 2:	2.1. Materials are precisely quantified according to the mixing ratio.	The required number of cement blocks is calculated			2

Fabricate cement blocks (25%)		The required number of cement pavers is calculated			2
	2.2. Materials are properly batched and mixed based on the design ratio	Mix ratio is respected			2
		Materials are batched			1
		Mixed material mortar/paste is homogeneous			2
		Mixed mortar is workable			1
		No impurities observed in mortar			1
		Procedures of mixing is respected			2
	2.3. Mixed materials are correctly poured and compacted into the mould according to the designed shape	Mixed materials are poured into the mould			2
		Process of pouring is respected			2
		Mixed materials are compacted			2
	2.4. Cement blocks is adequately de-moulded and cured according to the standard	Cement blocks is de-moulded			2
		No defects observed on demoulded blocks			2
		Cement blocks is cured			2
Learning outcome 3: Fabricate cement pavers (25%)	3.1. Materials are precisely quantified according to the mixing ratio	The required quantity of cement is calculated			2
		The quantity of sand is calculated			2
	3.2. Materials are conveniently batched and mixed based on the design ratio	Mix ratio is respected			2
		Materials are batched			1
		Mixed material paste is homogeneous			2
		Mixed material mortar is workable			1
		No impurities observed in mortar			1
		Procedures of mixing is respected			2
	3.3. Mixed materials are correctly poured and compacted into the mold according to the designed shape	Mixed materials are poured into the mould			2
		Process of pouring is respected			2
		Mixed materials are			2

		compacted			
	3.4. Pavers is adequately de-moulded and cured according to the standard.	Fabricated cement pavers are de-moulded			2
		No defects observed on fabricated cement pavers			2
		Fabricated cement pavers are cured			2
Learning outcome 4: Prepare ground base (10%)	4.1. The paving area is properly cleared and levelled in accordance to design/work specification and standard	Working area is levelled and set out			2
		Water drainage system is set			1
		Working area diagonal is checked			1
	4.2. Ground base is accurately investigated according to the work specifications	Soil type is checked			1
		Ground base is wet			1
	4.3. The laying place is conveniently compacted in accordance to the type of traffic to be supported, and work specifications or desired compaction rate	The laying place is compacted			3
		The laying place is levelled			1
Learning outcome 5: Lay pavers (30%)	5.1. Edge restraint is conveniently supported according to the standards	The edge of area to be paved is supported			3
		Precasted concrete kerbs are fixed			3
	5.2. Sand bedding is evenly spread along the ground base	Sand bedding is spread along the ground base			3
		Ground base is compacted			2
		Ground based is leveled			2
	5.3. Pavers are correctly applied according to the design	Starting point is selected			3
		Starting point is marked			2
		First paver is placed on the marked point			2
	5.4. Pavers’ Joints are correctly filled according to the design/work pacifications	All joints are filled			2
		Excess materials are removed			2
	5.5. Pavers’ Joints are properly sealed according to the design/work specifications	Paver’s joints are sealed			3
	5.6. Tools, equipment and work area are adequately cleaned according to standard.	Tools are cleaned			1
		Equipment are cleaned			1
		Work area is cleaned			1
Total marks		100			

Percentage Weightage	100%
Minimum Passing line % (Aggregate): 70%	

References:

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